



Question Bank (2)

Microbiology and Infection Control

(MED12114)

Spring Semester 2025/2026

Antimicrobial Resistance

- 1. Which enzyme destroys β -lactam antibiotics?**
 - A. Coagulase**
 - B. β -lactamase**
 - C. Catalase**
 - D. Oxidase**
- 2. Resistance to tetracycline may occur due to:**
 - A. Increased drug permeability**
 - B. Decreased drug permeability**
 - C. Increased antibody formation**
 - D. Increased drug absorption**
- 3. Which test uses antibiotic disks placed on agar?**
 - A. Broth dilution test**
 - B. Disk diffusion test**
 - C. Gram stain**
 - D. Blood culture**
- 5. A bacterium that changes its penicillin-binding proteins becomes resistant mainly to:**
 - A. Aminoglycosides**
 - B. β -lactams**
 - C. Sulpha drugs**

D. Chloramphenicol

6. Which example represents non-genetic resistance?

A. Plasmid transfer of β -lactamase gene

B. Mutation in PBPs

C. Dormancy in tuberculosis bacteria

D. Transposon-mediated resistance

7. In disk diffusion testing, a large clear zone around an antibiotic disk usually means the organism is:

A. Resistant

B. Susceptible

C. Contaminated

D. Dormant

8. A nurse should monitor renal and hearing function when a patient receives:

A. Aminoglycosides

B. Tetracycline

C. Sulpha drugs

D. Erythromycin

9. A patient asks why a sensitivity test is needed before antibiotics. The best nursing response is:

A. "It helps choose the most effective antibiotic".

B. "It prevents all allergic reactions".

C. "It confirms viral infection".

D. "It replaces the need for culture".

10. A nurse receives a report showing bacteria resistant to antibiotic B in disk diffusion. The nurse should expect the physician to:

A. Continue antibiotic B

B. Select another effective antibiotic

C. Stop all antibiotics permanently

D. Ignore the result

11.A child is prescribed tetracycline. Which complication should the nurse consider?

A. Greyish discoloration of teeth

B. Nephrotoxicity only

C. Aplastic anemia

D. Severe hypoglycemia

12.Which nursing action helps limit antimicrobial resistance?

A. Encourage stopping antibiotics when symptoms improve

B. Use antibiotics without sensitivity testing

C. Maintain correct drug levels throughout therapy

D. Give antibiotics for all fevers

13.A patient's infection does not respond to penicillin because the organism lacks a cell wall. Which organism is most likely involved?

A. Mycoplasma

B. Staphylococcus aureus

C. Streptococcus pyogenes

D. Escherichia coli

14.A bacterial isolate resists chloramphenicol due to alteration of the 50S ribosomal subunit. This is an example of:

A. Drug inactivation

B. Altered receptor/target site

C. Decreased drug permeability

D. Metabolic inactivity

15.In broth dilution testing, bacterial growth is inhibited at 2 µg/mL but present at 1 µg/mL. The MIC is:

A. 1 µg/mL

B. 2 µg/mL

C. 4 µg/mL

D. 8 µg/mL

16. Early antibiotic use may suppress antibody formation. This can lead to:

A. Complete immunity

B. Relapse of disease

C. Tooth discoloration

D. Increased drug permeability

17. Which prescription order should the nurse question?

A. Antibiotic after culture and sensitivity test

B. Antibiotic maintained for full prescribed course

C. Broad antibiotic given without indication or sensitivity test

D. Monitoring for drug toxicity

18. A patient on chloramphenicol develops fatigue and pallor. Which complication should the nurse suspect?

A. Aplastic anemia

B. Ototoxicity

C. Tooth discoloration

D. Superinfection only

19. A patient develops oral Candida after prolonged antibiotic therapy. The most likely explanation is:

A. Suppression of normal bacterial flora

B. Increased antibody production

C. β-lactamase production

D. Loss of bacterial capsule

20. Which is the best approach to reduce antibiotic resistance in hospital practice?

A. Use the newest antibiotic for every infection

B. Avoid antibiotics completely

C. Use antibiotics according to sensitivity results

D. Give low doses to reduce toxicity

21. Which patient teaching statement best promotes safe antibiotic use?

A. "Stop the drug once fever disappears".

B. "Share leftover antibiotics with family members".

C. "Complete the full course as prescribed".

D. "Use antibiotics for viral colds".

22. A nurse is preparing an education poster on antimicrobial resistance. Which title is most appropriate?

A. "Antibiotics Cure All Infections"

B. "Use Antibiotics Wisely: Test, Treat, Complete"

C. "Stop Antibiotics Early"

D. "Avoid Laboratory Tests"

23. Which nursing care plan is most appropriate for a patient receiving aminoglycosides?

A. Monitor kidney function and hearing changes

B. Monitor only blood glucose

C. Encourage skipping doses

D. Avoid asking about allergies

24. A nurse creates a checklist to prevent resistance. Which item should be included?

A. Start antibiotics before assessment in all cases

B. Use antibiotics without culture results

C. Check sensitivity results before antibiotic selection when possible

D. Stop antibiotics once symptoms improve?

Antimicrobial Chemotherapy

1. Antimicrobial drugs are substances that:

A. Stimulate bacterial growth

B. Kill or inhibit the growth of microorganisms

- C. Affect viruses only
- D. Increase bacterial resistance

2. Antibiotics classified according to source may be:

- A. Bacteriostatic and bactericidal
- B. Natural and synthetic
- C. Broad and narrow
- D. Toxic and non-toxic

3. Bactericidal antibiotics:

- A. Stop bacterial multiplication only
- B. Kill bacteria
- C. Stimulate immunity
- D. Act on fungi only

4. Bacteriostatic antibiotics:

- A. Kill bacteria immediately
- B. Prevent bacteria from multiplying
- C. Destroy bacterial DNA only
- D. Cause cell wall rupture only

5. Broad-spectrum antibiotics act against:

- A. One type of bacteria only
- B. A limited group of bacteria
- C. A larger group of bacteria
- D. Viruses only

6. Narrow-spectrum antibiotics are useful because they:

- A. Act against a limited group of bacteria
- B. Kill all microorganisms
- C. Are always toxic
- D. Are used randomly

7. β -lactam antibiotics act mainly by inhibiting:

- A. Protein synthesis
- B. Cell wall synthesis
- C. Cell membrane function
- D. RNA synthesis only

8. Bacterial cell wall synthesis is important because it involves:

- A. Peptidoglycan formation
- B. Viral replication
- C. Antibody production
- D. Fungal capsule formation

9. β -lactams bind to:

- A. DNA polymerase
- B. Penicillin-binding proteins
- C. 70S ribosome only
- D. Cell membrane lipids

10. Inhibition of transpeptidase leads to:

- A. Stronger bacterial cell wall
- B. Cell death and lysis
- C. Increased bacterial growth
- D. Increased protein synthesis

11. A nurse is caring for an immunocompromised patient with bacterial infection. Which type of antibiotic is preferred?

- A. Bacteriostatic
- B. Bactericidal
- C. Antiviral
- D. Antifungal only

12.A patient receives penicillin. The nurse understands that penicillin acts best on:

- A. Dormant bacterial cells
- B. Growing metabolically active bacterial cells
- C. Human ribosomes
- D. Viruses

13.A patient is prescribed ciprofloxacin. The nurse knows this drug interferes with:

- A. DNA replication
- B. Cell wall receptors
- C. Folic acid synthesis
- D. Cell membrane permeability only

14.Rifampicin works mainly by inhibiting:

- A. RNA synthesis
- B. Cell wall synthesis
- C. Protein synthesis
- D. Cell membrane function

15.A nurse is teaching a patient about antibiotic use. Which statement is correct?

- A. "Use antibiotics randomly when you feel ill".
- B. "Antibiotics should be used according to culture results when possible".
- C. "Broad-spectrum antibiotics are always the best choice".
- D. "Antibiotics are effective against all viruses".

16.A patient develops toxicity after receiving Polymyxin B. Which adverse effects should the nurse monitor for?

- A. Nephrotoxicity and neurotoxicity
- B. Tooth discoloration only
- C. Hyperglycemia
- D. Hearing improvement

17.A drug changes cell membrane permeability and causes leakage of essential metabolic substances. This drug acts by:

- A. Inhibition of cell wall synthesis
- B. Inhibition of cell membrane function
- C. Inhibition of nucleic acid synthesis
- D. Inhibition of folic acid synthesis

18.An antibiotic interferes with the 70S bacterial ribosome. Its main action is:

- A. Inhibition of protein synthesis
- B. Inhibition of DNA replication
- C. Cell membrane disruption
- D. Cell wall synthesis stimulation

19.Aminoglycosides act mainly on which ribosomal subunit?

- A. 30S subunit
- B. 50S subunit
- C. 80S subunit
- D. 60S subunit

20.Chloramphenicol acts mainly on which ribosomal subunit?

- A. 30S subunit
- B. 50S subunit
- C. 80S subunit
- D. Cell wall receptor

21.A nurse reviews a prescription for broad-spectrum antibiotics without culture results. What is the best nursing action?

- A. Administer immediately without concern
- B. Question the order and clarify if culture is needed
- C. Stop all treatment
- D. Increase the dose

22. Which strategy is best to reduce antibiotic resistance?

- A. Use antibiotics randomly
- B. Use broad-spectrum antibiotics always
- C. Use antibiotics based on culture results
- D. Increase antibiotic dose for all patients

23. Overuse of broad-spectrum antibiotics may lead to:

- A. Decreased resistance
- B. Increased resistance
- C. No effect on bacteria
- D. Faster recovery in all cases

24. Which antibiotic choice is more appropriate when the causative bacteria is known?

- A. Narrow-spectrum antibiotic
- B. Random antibiotic
- C. Broad-spectrum antibiotic always
- D. Antiviral drug

25. Which nursing judgment is correct regarding bacteriostatic antibiotics?

- A. They kill bacteria directly
- B. They prevent bacterial multiplication
- C. They destroy the cell membrane only
- D. They are used only for viruses

26. A nurse is preparing a patient education plan about antibiotics. Which instruction should be included?

- A. Take antibiotics only when prescribed
- B. Stop antibiotics after one dose
- C. Share antibiotics with family members
- D. Use broad-spectrum antibiotics for all infections

27. A nurse designs a checklist for safe antibiotic administration. Which item is most important?

- A. Check allergy history before giving antibiotics
- B. Give antibiotics without assessment
- C. Ignore culture results
- D. Encourage missed doses

28. A nurse prepares an educational poster to reduce antibiotic resistance. Which title is best?

- A. "Use Antibiotics Randomly"
- B. "Broad-Spectrum Antibiotics for Everyone"
- C. "Use Antibiotics Wisely Based on Culture Results"
- D. "Antibiotics Cure All Diseases"

29. A nursing care plan for a patient receiving Polymyxin B should include:

- A. Monitoring kidney and neurological status
- B. Encouraging self-medication
- C. Ignoring side effects
- D. Avoiding patient education

30. A nurse is planning teaching about protein synthesis inhibitors. Which information should be included?

- A. They act on bacterial 70S ribosomes
- B. They act only on human 80S ribosomes
- C. They stimulate bacterial multiplication
- D. They increase folic acid production

1-Case Study: Choosing an Antibiotic Type

A 65-year-old immunocompromised patient is admitted with a severe bacterial infection. The physician prescribes a bactericidal antibiotic.

Questions:

Why is a bactericidal antibiotic preferred in this patient?

What is the main action of bactericidal antibiotics?

What nursing assessments are important before administering the antibiotic?

2.Case Study: Bacteriostatic Antibiotic

A patient with a mild bacterial infection is prescribed a bacteriostatic antibiotic.

Questions:

How does a bacteriostatic antibiotic affect bacteria?

How is it different from a bactericidal antibiotic?

What should the nurse teach the patient about completing the antibiotic course?

3.Scenario: Broad-Spectrum Antibiotic Use

A patient is started on a broad-spectrum antibiotic before culture results are available.

Questions:

What does broad-spectrum antibiotic mean?

What is one risk of overusing broad-spectrum antibiotics?

What should the nurse check when culture results become available?

4.Scenario: Narrow-Spectrum Antibiotic

A culture report identifies a specific causative bacterium. The physician changes the antibiotic from broad-spectrum to narrow-spectrum.

Questions:

What does narrow-spectrum antibiotic mean?

Why may narrow-spectrum antibiotics be preferred when the organism is known?

How can this practice help reduce antibiotic resistance

Fungi and Fungal infections .

1.Fungi are classified as:

- A. Prokaryotic organisms
- B. Eukaryotic organisms
- C. Viruses
- D. Non-cellular organisms

2.The fungal cell wall is mainly composed of:

- A. Peptidoglycan
- B. Chitin and polysaccharides
- C. Cholesterol
- D. Lipopolysaccharide

3.The sterol present in fungal cell membrane is:

- A. Cholesterol
- B. Ergosterol
- C. Peptidoglycan
- D. Chitin

4.Human cell membranes contain:

- A. Ergosterol
- B. Chitin
- C. Cholesterol
- D. Peptidoglycan

5.Candida albicans is classified morphologically as:

- A. Yeast
- B. Hyphae
- C. Virus
- D. Bacterium

6.Fungi are insensitive to antibacterial antibiotics because:

- A. They lack cytoplasm
- B. They have a different cell wall composition
- C. They have circular DNA
- D. They are prokaryotic

7.Yeasts reproduce asexually by:

- A. Binary fission

- B. Budding
- C. Fragmentation
- D. Conjugation

8. Hyphae are:

- A. Single round fungal cells
- B. Tubular projections that form mycelium
- C. Bacterial spores
- D. Viral particles

9. Dimorphic fungi can exist as:

- A. Bacteria and viruses
- B. Yeast or hyphal form
- C. Gram-positive and Gram-negative forms
- D. Spores only

10. A dimorphic fungus appears as yeast form mainly:

- A. At 22°C
- B. At 37°C or in tissue
- C. In spoiled grains only
- D. In air only

11. A nurse is caring for a patient receiving long-term antibiotics who develops oral white patches. The most likely cause is:

- A. Bacterial infection
- B. Viral infection
- C. Fungal overgrowth
- D. Autoimmune disease

12. A patient presents with fungal infection affecting skin, hair, and nails. This is classified as:

- A. Systemic mycosis
- B. Subcutaneous mycosis

- C. Superficial mycosis
- D. Opportunistic infection only

13. A farmer develops chronic foot swelling after injury from a thorn, with sinus formation. The most likely diagnosis is:

- A. Oral candidiasis
- B. Mycetoma
- C. Viral infection
- D. Lung infection

14. A neonate develops Candida infection after delivery. The possible source is:

- A. Cross infection from mother to neonate during delivery
- B. Air only
- C. Insect bite
- D. Spoiled peanuts

15. A patient with oral thrush has white adherent patches on the tongue and inner cheeks. The likely diagnosis is:

- A. Leukoplakia
- B. Oral candidiasis
- C. Aphthous ulcer
- D. Herpes simplex

16. Aflatoxin is produced by:

- A. Candida albicans
- B. Aspergillus flavus
- C. Dermatophytes
- D. Penicillium only

17. Aflatoxin is associated with:

- A. Lung fibrosis
- B. Hepatocellular carcinoma

- C. Renal failure
- D. Diabetes mellitus

18. Systemic fungal infections spread mainly through:

- A. Skin contact only
- B. Bloodstream, hematogenous spread
- C. Hair follicles only
- D. Food contamination only

19. A systemic fungal infection may affect:

- A. Hair only
- B. Nails only
- C. Brain, lungs, and blood
- D. Skin surface only

20. A patient with AIDS is at high risk for Candida infection because of:

- A. Strong immunity
- B. Immunocompromised state
- C. High blood pressure
- D. Normal flora balance only

21. Which patient has the highest risk for candidiasis?

- A. Healthy adult with no medication use
- B. Patient receiving prolonged broad-spectrum antibiotics
- C. Patient with mild headache
- D. Patient with simple muscle strain

22. Which nursing action is most important for a patient receiving Amphotericin B?

- A. Monitor kidney function
- B. Encourage self-medication
- C. Stop all fluids
- D. Ignore adverse effects

23. Amphotericin B use is limited mainly because it may cause:

- A. Nephrotoxicity
- B. Tooth discoloration
- C. Hypertension only
- D. Hyperthyroidism

24. Which antifungal drug group inhibits ergosterol synthesis?

- A. Azoles
- B. Penicillins
- C. Cephalosporins
- D. Aminoglycosides

25. Which finding should the nurse recognize as oral candidiasis?

- A. White adherent patches on buccal mucosa
- B. Black stool
- C. Dry cough only
- D. Yellow skin discoloration

26. A nurse is preparing health education for a patient with oral candidiasis. Which instruction is appropriate?

- A. Maintain good oral hygiene
- B. Stop treatment once patches appear
- C. Share antifungal medication with others
- D. Use antibiotics randomly

27. A nurse designs a prevention plan for obese patients at risk of intertrigo. Which advice should be included?

- A. Keep skin folds clean and dry
- B. Keep skin folds moist all the time
- C. Avoid hygiene care
- D. Use antibiotics without prescription

28. A nurse prepares teaching for a diabetic patient to prevent Candida infection. Which point is most appropriate?

- A. Control blood sugar and maintain hygiene
- B. Ignore oral lesions
- C. Use broad-spectrum antibiotics frequently
- D. Avoid reporting symptoms

29. A nursing care plan for a patient receiving topical Nystatin should include:

- A. Apply to affected skin or mucous membranes as prescribed
- B. Use only for systemic bloodstream infection
- C. Monitor for nephrotoxicity as the main effect
- D. Stop after one dose

30. A nurse prepares an infection-control plan in a nursery to prevent Candida cross infection. Which measure is most appropriate?

- A. Hand hygiene and preventing infant-to-infant spread
- B. Sharing feeding equipment
- C. Ignoring oral white patches
- D. Avoiding cleaning procedures

Case 1: Oral White Patches

A nurse observes white, adherent patches on the tongue and inner cheeks of a patient receiving long-term broad-spectrum antibiotics.

What is the most likely diagnosis?

What is the main predisposing factor in this case?

Which organism is commonly responsible for this condition?

What nursing care should be provided for this patient?

Case 2: Neonatal Candida Infection

A premature neonate in the nursery develops signs of Candida infection.

Why are premature neonates more susceptible to Candida infection?

How can Candida cross infection occur in neonates?

What infection-control measures should nurses apply in the nursery?

Why is early detection important?

Case 3: Diabetic Patient with Oral Thrush

A diabetic patient complains of soreness in the mouth. On assessment, the nurse finds white patches on the buccal mucosa and tongue.

What is the likely fungal infection?

How does diabetes increase the risk of this infection?

Which sites are commonly affected in oral candidiasis?

What health education should the nurse provide?

Case 4: Denture-Associated Infection

An elderly patient who wears dentures develops redness and discomfort under the denture area.

What Candida-associated oral lesion may be present?

Why can dentures increase the risk of candidiasis?

What nursing advice should be given about denture care?

Which type of antifungal therapy may be used for superficial mucosal infection?

Answers

Case 1: Oral White Patches

.The most likely diagnosis is oral candidiasis, also called oral thrush

The main predisposing factor is long-term broad-spectrum antibiotic use, which
.disrupts normal flora

.The common organism is *Candida albicans*

Nursing care includes assessing the oral cavity, maintaining good oral hygiene, administering prescribed antifungal therapy such as nystatin, and educating the
.patient to complete treatment

Case 2: Neonatal Candida Infection

Premature neonates are more susceptible because their immunity is weak and
.immature

Candida cross infection can occur from mother to neonate during delivery or .between infants in nurseries

Nurses should apply hand hygiene, clean equipment, avoid sharing .contaminated items, and monitor infants for early signs of infection

Early detection is important because neonates can develop serious infection .quickly

Case 3: Diabetic Patient with Oral Thrush

.The likely infection is oral candidiasis / oral thrush

Diabetes increases the risk because poor health status and high glucose levels .encourage Candida overgrowth

Common sites include the buccal mucosa, inner cheeks, tongue, roof of the .mouth, and throat

The nurse should teach good oral hygiene, blood glucose control, reporting oral .lesions early, and completing antifungal treatment

Case 4: Denture-Associated Infection

.The patient may have denture stomatitis

Dentures increase risk because they create local irritation and a moist area .where Candida can grow

The nurse should advise proper denture cleaning, removing dentures when .appropriate, maintaining oral hygiene, and reporting redness or soreness

Topical antifungal therapy, such as nystatin, may be used for superficial mucosal infection.

Viruses and viral infections

1. Viruses are best described as:

- A. Independent living cells
- B. Multicellular organisms
- C. Obligate intracellular parasites
- D. Free-living bacteria

2. Viruses contain:

- A. DNA only
- B. RNA only
- C. Either DNA or RNA
- D. DNA and RNA together

3. The protein coat surrounding viral nucleic acid is called:

- A. Envelope
- B. Capsid
- C. Ribosome
- D. Cell wall

4. The capsid is made up of subunits called:

- A. Spores
- B. Capsomers
- C. Ribosomes
- D. Flagella

5. The viral envelope is:

- A. A peptidoglycan layer
- B. A lipoprotein coat
- C. A chitin wall
- D. A bacterial capsule

6. Viruses cannot reproduce outside a host cell because they:

- A. Lack genetic material
- B. Depend on host cell machinery
- C. Are too large
- D. Have bacterial ribosomes

7. The main function of viral nucleic acid is to:

- A. Produce antibodies

B. Carry genetic information for viral replication

C. Form bacterial cell walls

D. Destroy host ribosomes

8. The viral capsid helps the virus by:

A. Producing energy

B. Protecting viral nucleic acid

C. Making host antibodies

D. Acting as a nucleus

9. Enveloped viruses are classified according to:

A. Presence of mitochondria

B. Presence of lipid envelope

C. Presence of bacterial cell wall

D. Presence of cytoplasm

10. Glycoprotein spikes help viruses to:

A. Attach to host cell receptors

B. Produce ATP

C. Form ribosomes

D. Digest bacterial cell wall

11. A nurse is handling a virus that is easily destroyed by detergents and drying. This virus is most likely:

A. Non-enveloped virus

B. Enveloped virus

C. Bacterial cell

D. Spore-forming organism

12. A virus attaches to a host cell using surface proteins. Which structure is responsible?

A. Nucleic acid

B. Glycoprotein spikes

C. Ribosomes

D. Cell wall

13. A patient with influenza should be managed mainly using:

A. Droplet precautions

B. Blood precautions only

C. No precautions

D. Food precautions only

14. A patient with COVID-19 has shortness of breath. Which nursing action is important?

A. Monitor oxygen saturation

B. Ignore respiratory signs

C. Give antibiotics only

D. Stop isolation precautions

15. A patient with hepatitis B should be cared for using:

A. Standard precautions

B. Airborne precautions only

C. No precautions

D. Droplet precautions only

16. A patient develops flaccid paralysis after a viral infection. Which mechanism best explains this condition?

A. Liver cell destruction

B. Skin inflammation

C. Destruction of motor neurons

D. Immune suppression only

17. Poliovirus primarily affects:

A. Liver cells

B. Skin

C. Motor neurons in the CNS

D. Kidney cells

18. A child presents with fever and swelling of the parotid gland. The most likely virus is:

A. Measles virus

B. Rubella virus

C. Mumps virus

D. Poliovirus

19. Measles is considered highly contagious because it spreads through:

A. Blood only

B. Airborne droplets

C. Food contamination

D. Insect bites

20. A pregnant woman in the first trimester is exposed to a viral infection. The fetus develops deafness and cardiac defects. Which virus is responsible?

A. Measles virus

B. Rubella virus

C. Mumps virus

D. Poliovirus

21. Which preventive measure is most effective against measles, mumps, and rubella?

A. Antibiotic therapy

B. MMR vaccination

C. Hand washing only

D. Isolation only

22. A nurse is planning infection-control measures for a patient with measles. Which is the best approach?

A. Contact precautions only

B. Airborne precautions

C. Blood precautions only

D. No precautions needed

23. Which nursing action is most appropriate for a patient with suspected COVID-19?

A. Use PPE and isolation precautions

B. Place the patient in a shared room without precautions

C. Ignore oxygen saturation

D. Encourage visitors without masks

24. A patient with HIV/AIDS needs nursing care. Which action is most appropriate?

A. Apply standard precautions and provide emotional support

B. Avoid all contact with the patient

C. Use antibiotics as prevention for all viral infections

D. Ignore recurrent infections

25. Which case requires urgent nursing attention?

A. Mild cold symptoms with stable breathing

B. COVID-19 patient with low oxygen saturation

C. Patient asking about vaccination

D. Patient with mild body aches only

26. A nurse prepares health education for preventing viral infections. Which instruction should be included?

A. Avoid vaccination

B. Practice hand hygiene and take recommended vaccines

C. Use antibiotics for all viral infections

D. Stop isolation precautions early

27. A nurse designs a plan to prevent hepatitis B transmission. Which measure is most appropriate?

A. Safe injection practices and vaccination

B. Sharing needles

C. Avoiding vaccination

D. Using antibiotics after exposure

28. A nurse prepares a teaching plan for influenza prevention. Which point should be included?

A. Annual vaccination and respiratory hygiene

B. Avoid hand washing

C. Antibiotics prevent influenza

D. No need to monitor complications

Case-Based Questions

Case 1: Influenza Patient

A patient presents with fever, cough, and body aches. The physician suspects influenza.

What is the most likely causative virus?

What is the main route of transmission?

What complication should the nurse monitor for?

What nursing precautions should be applied?

What preventive measures should be taught to the patient?

Case 2: COVID-19 Patient

A patient presents with fever, cough, loss of smell, shortness of breath, and fatigue. COVID-19 is suspected.

What is the causative virus?

What are the main routes of transmission?

What important nursing assessment should be performed?

What infection-control precautions should be used?

What complications may occur?

Case 3: Measles Infection

A child presents with fever, cough, runny nose, red eyes, and skin rash. Measles is suspected.

What is the causative virus?

Why is measles considered highly contagious?

What type of isolation precaution is needed?

What complications may occur?

What vaccine prevents measles?

Health Care Associated Infections (HAIs) .

1. Nosocomial infections are also known as:

- A. Genetic infections
- B. Hospital-acquired infections
- C. Autoimmune infections
- D. Nutritional infections

2. Nosocomial infections usually develop:

- A. Immediately after admission
- B. 12 hours after admission
- C. 24 hours after admission
- D. 48 hours or more after admission

3. The most common pathogens causing nosocomial infections are:

- A. Viruses
- B. Fungi
- C. Bacteria
- D. Parasites

4. The term “nosocomial” originates from a Greek word meaning:

- A. Home
- B. Hospital
- C. Disease
- D. Patient

5. Which of the following is a source of nosocomial infection?

- A. Healthcare personnel
- B. Contaminated instruments

C. Patient's own endogenous flora

D. All of the above

6. Hospitalized patients are more susceptible to HAIs because:

A. They are always healthy

B. They may have weakened immunity and invasive procedures

C. They do not contact healthcare workers

D. They never use antibiotics

7. Long-term antibiotic therapy increases infection risk because it:

A. Strengthens normal flora

B. Disrupts normal microbial flora

C. Prevents opportunistic pathogens

D. Removes all hospital organisms

8. Vascular catheterization increases the risk of infection because it:

A. Protects the bloodstream

B. Provides a direct pathway for pathogens into the blood

C. Prevents bacterial growth

D. Reduces hospital flora

9. Mechanical ventilation increases infection risk mainly because:

A. It improves mucous membrane protection

B. It may damage the mucous membrane lining

C. It prevents microorganisms from entering

D. It has no relation to infection

10. Hospital flora can survive on:

A. Surfaces

B. Equipment

C. Linen and medical devices

D. All of the above

11. Before inserting a urinary catheter, the nurse should:

- A. Skip cleaning if the patient looks healthy
- B. Clean the periurethral area with antiseptic
- C. Insert the catheter without gloves
- D. Keep the catheter as long as possible

12. A nurse is using a stethoscope for different patients. What should be done between patients?

- A. Use it without cleaning
- B. Clean it before using it on another patient
- C. Share it only in ICU
- D. Clean it once daily only

13. A patient has an indwelling urinary catheter. Which nursing action helps prevent infection?

- A. Keep the catheter longer than needed
- B. Remove the catheter as soon as possible when no longer needed
- C. Disconnect the catheter frequently
- D. Ignore periurethral hygiene

14. A patient in ICU is colonized with MRSA. The best nursing action is to:

- A. Place the patient in isolation according to policy
- B. Move the patient close to other patients
- C. Stop hand hygiene
- D. Share equipment with other ICU patients

15. A patient with diabetes is admitted to hospital. Which nursing action helps reduce infection risk?

- A. Ignore blood glucose levels
- B. Control diabetes well
- C. Avoid hygiene care
- D. Increase unnecessary catheter use

16. A hospitalized elderly patient with a urinary catheter develops fever after 3 days. This is most likely:

- A. Community-acquired infection
- B. Nosocomial infection
- C. Genetic disease
- D. Autoimmune reaction

17. A patient develops pneumonia after being on mechanical ventilation. This is most likely:

- A. Ventilator-associated pneumonia
- B. Surgical site infection
- C. Gastroenteritis
- D. Puerperal fever

18. A patient develops infection after surgery at the incision site. This type of HAI is:

- A. Bloodstream infection
- B. Surgical site infection
- C. Urinary tract infection
- D. Ventilator-associated pneumonia

19. Unsafe injection practice may transmit which nosocomial viral infections?

- A. Hepatitis B and C
- B. MRSA only
- C. Candida albicans
- D. Aspergillus spp.

20. Overgrowth of the patient's own Candida albicans in hospital is usually related to:

- A. Strong immunity only
- B. Endogenous microflora with compromised immunity
- C. No antibiotic exposure
- D. Short hospital stay only

21. Which patient has the highest risk of developing nosocomial infection?

- A. Young healthy visitor
- B. Elderly ICU patient with vascular catheter and diabetes
- C. Healthy nurse after hand hygiene
- D. Patient admitted for 2 hours only

22. Which action is the best way to break the chain of infection in hospital?

- A. Hand hygiene before and after touching each patient
- B. Sharing equipment between patients
- C. Keeping urinary catheters for long periods
- D. Avoiding isolation for infected patients

23. A nurse finds that a urinary catheter is no longer needed. What is the best decision?

- A. Keep it for convenience
- B. Remove it as soon as possible
- C. Disconnect it from the drainage bag
- D. Replace it daily without indication

24. Which order should the nurse question to prevent nosocomial infection?

- A. Use the correct antibiotic for the specific germ
- B. Keep an unnecessary urinary catheter for several extra days
- C. Isolate infected ICU patients
- D. Clean non-critical equipment between patients

25. Which practice is most appropriate for antibiotic therapy in HAIs?

- A. Use any antibiotic randomly
- B. Use the right antibiotic for the specific germ
- C. Use antibiotics without diagnosis
- D. Stop antibiotics after one dose

26. A nurse is preparing an infection-control plan. Which intervention should be included first?

- A. Hand hygiene before and after patient contact
- B. Sharing stethoscopes without cleaning
- C. Avoiding PPE
- D. Ignoring contaminated linen

27. A nurse develops a prevention plan for catheter-associated UTI. Which point should be included?

- A. Use urinary catheters only when needed
- B. Keep urinary catheters for all patients
- C. Avoid periurethral cleansing
- D. Insert catheters without aseptic technique

28. A nurse prepares education for healthcare workers about MRSA and VRE. Which statement is correct?

- A. They are multidrug-resistant organisms
- B. They are harmless normal flora always
- C. They never spread in hospitals
- D. They are prevented by avoiding hand hygiene

29. A nurse is planning care for a patient at high risk of HAI. Which combined plan is best?

- A. Hand hygiene, aseptic technique, device care, and early catheter removal
- B. No isolation and shared equipment
- C. Random antibiotic use
- D. Prolonged invasive device use

Case 1: Fever After Hospital Admission

A 70-year-old patient develops fever and signs of infection 3 days after hospital admission. The infection was not present or incubating on admission.

What type of infection is this most likely to be?

Why is it considered a nosocomial infection?

Which patient factor increases this patient's risk?

What nursing actions are needed to prevent spread?

Case 2: Urinary Catheter Infection Risk

A patient has an indwelling urinary catheter inserted for several days and develops fever, suprapubic discomfort, and cloudy urine.

What type of nosocomial infection is suspected?

How does urinary catheterization increase infection risk?

What should the nurse do if the catheter is no longer needed?

What preventive measures should be followed before catheter insertion?

Good Luck

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